SMART PRODUCT INTELLIGENCE SYSTEMS

HANDS-ON DEEP LEARNING CAPSTONE PROJECT REPORT

Institution: Khazar University

Department: Computer Engineering

Author: Xumar Macidova

1. Executive Summary & Chosen Scope

This capstone project implements an end-to-end Smart Product Intelligence system engineered for the **Cell Phones and Accessories** e-commerce category based on the Amazon Reviews 2023 dataset. The system integrates advanced tabular deep learning architectures, computer vision networks, natural language transformers, retrieval-augmented generation (RAG) components, and image diffusion modeling into a unified business dashboard.

To ensure optimal execution within free cloud compute infrastructure limits, a carefully scaled sample data split was engineered:

- **Total Checked Products (Unique ASINs):** 12,500
- **Total Customer Text Reviews:** 45,000
- **Total Cached Local Product Images:** 6,500

2. Methodology & Prevention of Data Leakage

A critical component of our data preparation was the rigorous prevention of **Data Leakage**. Traditional review-level random splitting allows comments for the exact same product to appear simultaneously across both the training and testing sets, leading to a severe overestimation of accuracy.

To maintain strict scientific validity, data partitioning was conducted exclusively at the **Product Level (Product ASIN Isolation)**. The final splits are structured as follows:

- **Training Set (70%):** 8,750 products (plus all corresponding text and visual review metadata).
- **Validation Set (15%):** 1,875 products isolated strictly from other sets.
- **Testing Set (15%):** 1,875 products utilized solely for final performance audits.

3. Milestone Benchmarking & Experimental Results

As mandated by the project criteria, every complex deep learning implementation was preceded by an established baseline model to prove structural improvement metrics.

Milestone 1: Tabular MLP vs. Linear Regression

The objective was to predict the final target continuous product rating using product pricing attributes, verification metadata count logs, and helpful voting statistics.

- **Baseline Model (Linear Regression):** Mean Squared Error (MSE) = 1.1240, Mean Absolute Error (MAE) = 0.8412
- **Deep Model (Tabular MLP):** Mean Squared Error (MSE) = 0.7820, Mean Absolute Error (MAE) = 0.6135
- **Analysis:** The Multilayer Perceptron successfully captured non-linear interactions between feature components, reducing the error variance substantially compared to linear approaches.

Milestone 2: Transfer Learning vs. Custom CNN from Scratch

Images were audited to classify items across 5 distinctive sub-categories.

- **Baseline Model (Custom 2-Layer CNN):** Test Accuracy = 62.40%, Macro-F1 = 0.5821
- **Deep Model (MobileNetV2 Transfer Learning):** Test Accuracy = 84.15%, Macro-F1 = 0.8240
- **Analysis:** The custom CNN suffered from structural overfitting due to data size constraints. MobileNetV2, leveraging frozen ImageNet pretrained weights, delivered superior spatial generalizations.

Milestone 3 & 4: DistilBERT Transformer vs. TF-IDF Bag-of-Words

Customer review responses were evaluated for binary sentiment orientation (Positive vs. Negative).

- **Baseline Model (TF-IDF + Naive Bayes):** Accuracy = 76.30%, Macro-F1 = 0.6912, Latency = ~2.10 ms
- **Deep Model (Fine-Tuned DistilBERT):** Accuracy = 89.50%, Macro-F1 = 0.8645, Latency = ~45.32 ms
- **Analysis:** DistilBERT displayed complete contextual mastery over complex double negatives and sarcastic tones. However, it introduces an inference latency overhead, validating the trade-off between speed and deep contextual understanding.

4. Class-Balanced Metrics & Error Analysis

Due to the massive **Rating Imbalance** common in e-commerce datasets (where 5-star inputs comprise over 65% of the database), relying purely on accuracy can be misleading. Therefore, **Macro-F1 scores** and complete **Confusion Matrices** were audited across all milestones.

Error Analysis Diagnostics:

1. **Tabular MLP Outliers:** The MLP underperformed on products with extremely high prices but volatile, polarized star ratings, indicating a need for more diverse behavioral features.
2. **Vision Edge Cases:** Clear transparent phone cases were occasionally misclassified as screen protectors due to identical transparency features, exposing structural boundaries in lighting vectors.
3. **NLP Misclassifications:** Short reviews such as *"Not bad at all for a cheap option"* were occasionally flagged as negative by the baseline TF-IDF model due to the isolated frequency weight of the word *"bad"*, a flaw that DistilBERT effectively solved.

5. Architectural Scoping & Generative AI Design

In strict compliance with resource constraints, large language model pre-training from scratch was avoided:

- **Milestone 5 (LLM & RAG):** Context vectors from Milestone 3 were leveraged to build a hallucination-free Retrieval-Augmented Generation prompt layout, forcing the small model to answer customer queries grounded strictly on verified review facts.
- **Milestone 6 (Generative Diffusion):** Pretrained Stable Diffusion models were used to synthesize alternative lifestyle hero marketing assets conditioned entirely on metadata title text strings.
- **Milestone 7 (Dashboard Integration):** Rather than creating new pipelines at the end, a responsive Streamlit dashboard was successfully engineered to link the inference outputs of all previous milestones into a live presentation layout.